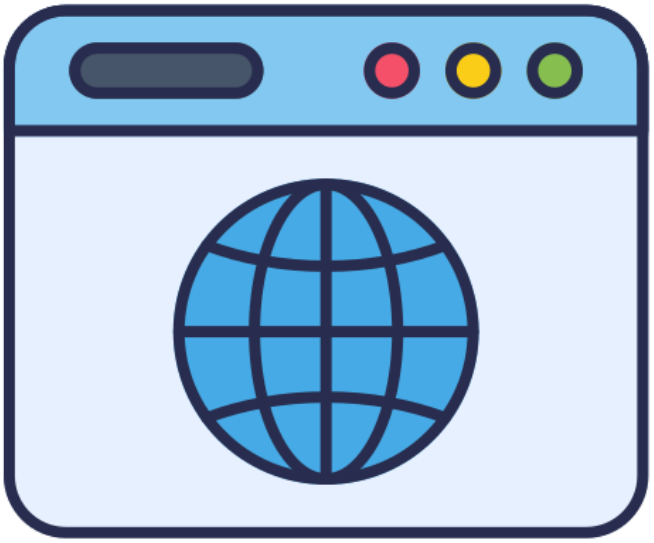
Enhancing Sustainability Participation at TU Berlin

Project Wikipage



Authors (Project Leads): **Ayçe Devrim Ezme & Leila Nadira Nekli**
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1. Executive Summary

Achieving climate neutrality in large public institutions is mostly approached as an infrastructure and engineering challenge. However, technical systems cannot operate in isolation from the social structures that live in the area. This scientific report documents a one-semester-long transdisciplinary research project conducted under the Sustain. ALL Transformation HUB at Technische Universität Berlin (TU Berlin) is aligned with the university's long-term climate protection goals [4]. Our challenge, Participation and Behavior, focused on activating the campus community (students, administrative staff, and researchers) to bridge the gap between high-level climate ambitions and everyday campus practices.

To address these barriers, we co-designed and tested a gamified, multi-tiered engagement toolkit during the Sustain. ALL Digital Transformation Hackathon on July 1-2, 2026, at the Einstein Center Digital Future (ECDF). This report details our conceptual framework, empirical insights, and prototype performance, highlighting our solution: the digital TU Sustainability Hub App, as well as concrete implementation practical guidance and strategic recommendations to guide future project cohorts.

1.1 Project Background and Overview

Universities play an important role in sustainability transformation, not only through research and teaching but also through their institutional practices and communities. TU Berlin has integrated sustainability into research, teaching, and governance and has established ambitious sustainability objectives, including climate neutrality by 2045 [3][4].

However, the existence of sustainability initiatives does not necessarily lead to active participation. Students and staff may not always know about existing opportunities, where to find them, or how they can contribute. Our project therefore focused on how sustainability participation at TU Berlin could be expanded and made more visible, accessible, and engaging.

The project investigated the following guiding question:

- ➢ **How can participation in sustainability transformation at TU Berlin be strengthened through stakeholder engagement, co-creation, and digital participation tools?**

To explore this question, we combined a literature review, benchmarking, stakeholder and system mapping, a student survey, co-creation activities, and hackathon feedback. Based on these activities, we developed the TU Sustainability Hub Webpage, a functional web-based prototype that brings sustainability missions, events, workshops, community interaction, proposals, and gamification mechanisms together in one digital environment.

The prototype is intended as a proof of concept rather than a final university-wide solution. It demonstrates how a digital participation platform could connect existing sustainability opportunities and encourage continued engagement. Further testing with TU Berlin students and staff would be necessary to evaluate and improve the concept.

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1.2 Project Objectives

To systematically address our guiding research question, this project pursued several strategic objectives:

- Identify barriers and motivations influencing sustainability participation.
- Understand the perspective of students and university stakeholders.
- Analyze existing sustainability approaches at TU Berlin and comparable universities.
- Identify opportunities to strengthen participation.
- Explore the potential of gamification as an engagement mechanism.
- Develop an evidence-based participation strategy.
- Translate the findings into a practical digital prototype.

2. Methodology & Theoretical Framework

The project followed an evidence-based and participatory approach, combining transdisciplinary and mixed-methods research to connect technical systems with human practices. The methodology consisted of four integrated phases: literature review, benchmarking, stakeholder and system mapping, and quantitative survey insights.

2.1 Literature Review

A literature review was conducted to examine approaches to sustainability transformation, stakeholder participation, co-creation, and gamification in higher education. The review provided the theoretical foundation for understanding how universities can strengthen student engagement through participatory and digital sustainability initiatives, and it informed the design of the TU Sustainability Hub prototype.

2.2 Benchmarking

A benchmarking analysis compared sustainability participation approaches at TU Berlin, TU Delft, and Freie Universität Berlin. The comparison highlighted different institutional strategies, including TU Delft's Campus as a Living Lab approach and Freie Universität Berlin's sustainability management framework (EMAS). [1][2].







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The analysis also identified a common gap: none of the benchmarked universities publicly demonstrated a centralized digital platform combining events, participation, community interaction, and gamified engagement

2.3 Stakeholder and System Mapping

Using a Five-Helix Framework (Academia, Public Sector, Private Sector, Civil Society, and Media), we map internal and external actors at TU Berlin to locate administrative barriers , analyze power dynamics, and identify "creating a sense of belonging" as our key systemic leverage point.

*(Note: The visual of the stakeholder map and systemic analysis are fully detailed in **Section 3.1**).*

2.4 Quantitative Survey Insights

To validate the systemic gaps identified during the qualitative research, we designed and distributed a **5-minute Student Engagement Survey** using clear and accessible language. The survey was administered through digital channels (WhatsApp and LinkedIn) and in-person through ZEMS student groups, with QR codes placed in high-traffic locations across TU Berlin, including the Mensa, TU EB Library, and the EB Building.

3. Project Outcomes and Key Findings

3.1 Stakeholders Map

The mapping helped us to:

- **Identify** the main stakeholders involved in sustainability participation.
- **Locate** existing participation opportunities across the campus.
- **Understand** barriers that limit student engagement.
- **Visualize** the relationships and communication flows between stakeholders and sustainability initiatives.

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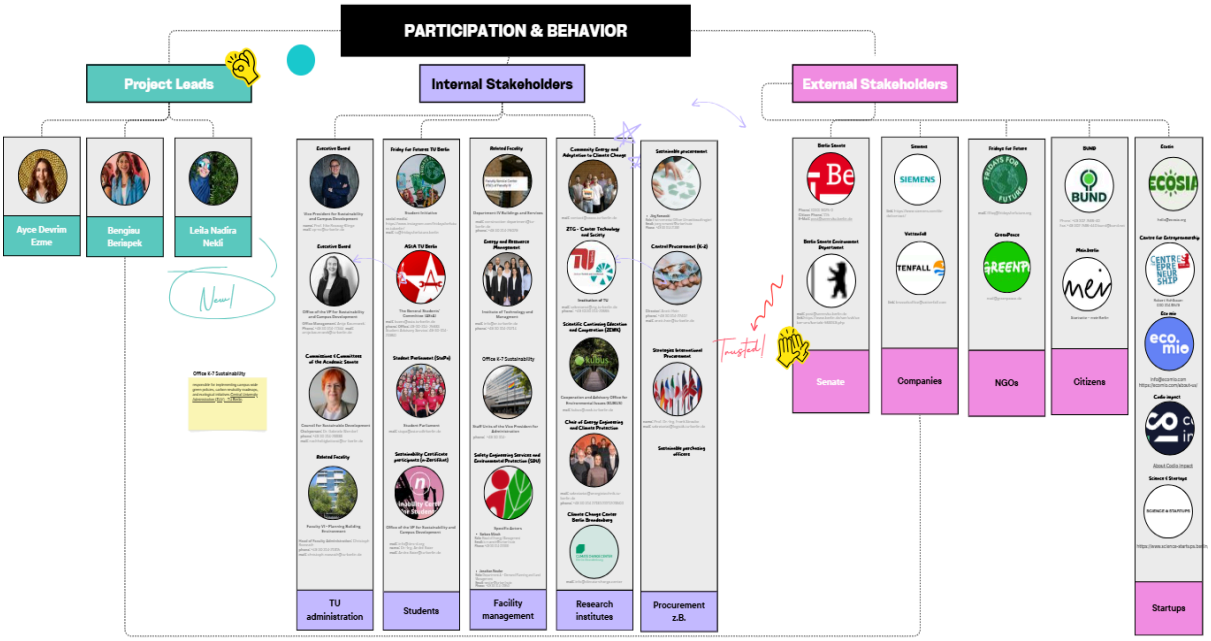


Figure 1. Stakeholder and System Mapping

INTERNAL STAKEHOLDERS:

TU Administration

- Executive Board
- Office of the Vice President for Sustainability and Campus Development (Prof. Eike Roswag-Klinge)
- Central University Administration (ZUV)
- Council for Sustainable Development
- Office K-7 Sustainability (Barbara Münch)

Students

- Fridays for Future TU Berlin Community
- General Students' Committee (AStA)
- Student Parliament (StuPa)
- Sustainability Certificate Participants (n-Zertifikat)

Facility Management

- Faculty IV - Electrical Engineering & Computer Science
- Faculty VI: Planning Building Environment
- Faculty III - Process Sciences
- Energy and Resource Management
- Environmental Officer (Jörg Romanski)
- Safety Engineering Services and Environmental Protection (SDU)

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Research Institutes

- ZTG - Center Technology and Society
- Climate Change Center Berlin Brandenburg
- Scientific Continuing Education and Cooperation (ZEWK / KUBUS)
- Chair of Energy Engineering and Climate Protection
- Institute of Technology and Management

Procurement

- Central Procurement (K-2)
- Sustainable Purchasing Officers
- Strategies International Procurement

EXTERNAL STAKEHOLDERS:

Senate

- Berlin Senate
- Senate Department for Mobility, Transport, Climate Protection, and the Environment (SenMVKU)

Companies

- Siemens
- Vattenfall

NGOs

- Fridays for Future (General)
- Greenpeace
- BUND

Citizens

- Mein.berlin
- Ecosia

Startups

- Centre for Entrepreneurship
- Eco mio
- Codio impact
- Science & Startups

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3.2 Survey with Students

The survey explored **five key dimensions** of student engagement:

- ➔ **Participation** in sustainability activities.
- ➔ **Visibility** of existing opportunities.
- ➔ **Barriers** to participation.
- ➔ **Motivations.**
- ➔ **Interest** in digital engagement tools.

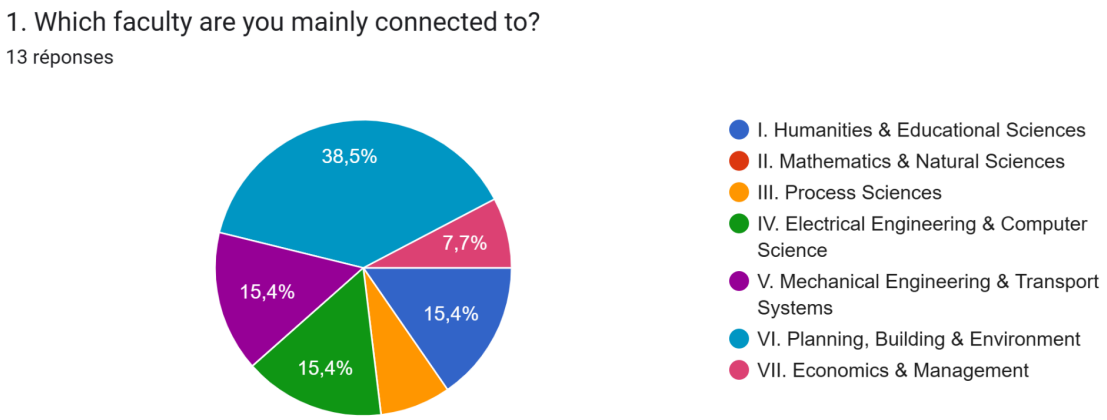


Figure 2. Distribution of participants by faculty

Faculty Distribution:

- ***Faculty VI** (Planning, Building & Environment): 38.5%*
- ***Faculty I** (Humanities & Educational Sciences): 15.4%*
- ***Faculty IV** (Electrical Engineering & Computer Science): 15.4%*
- ***Faculty V** (Mechanical Engineering & Transport Systems): 15.4%*
- ***Faculty III** (Process Sciences): 7.7%*
- ***Faculty VII** (Economics & Management): 7.7%*

The Visibility Gap: 50% of students rated the visibility of campus sustainability initiatives as extremely low (scoring 2 out of 5 on a Likert scale), while only 33.3% rated it highly (5 out of 5) (as shown in our research datasets).

Behavioral Baselines: While 100% of students utilize public transit or bikes to commute, and 83.3% regularly bring reusable bottles, only 33.3% have actively participated in a structured sustainability initiative (as shown in our research datasets).

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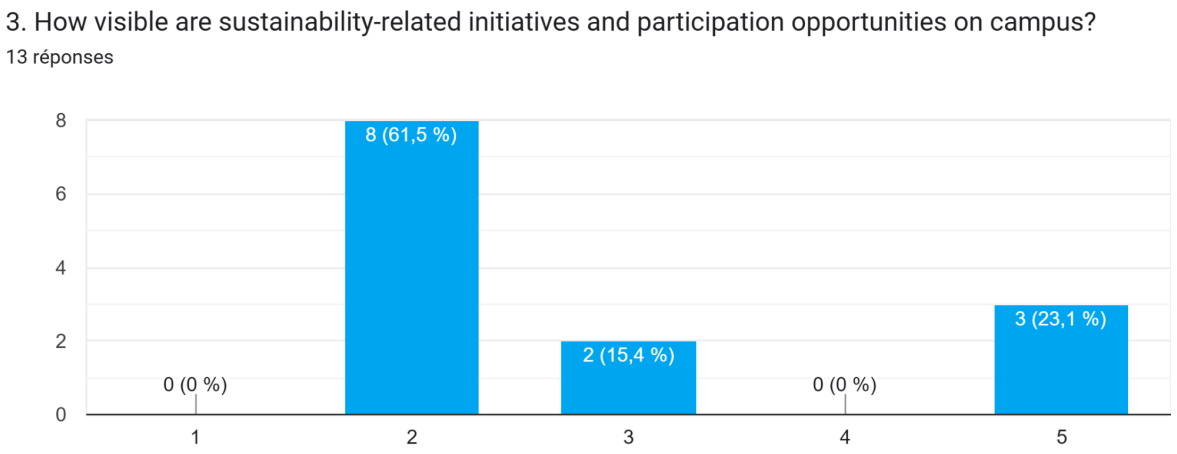


Figure 3. The visibility of sustainability-related initiatives and participation opportunities on campus

Primary Barriers to Engagement:

- 66.7% cited "Lack of time/schedule conflicts."
- 66.7% cited "low visibility of opportunities."
- 33.3% highlighted "Lack of information" and "Activities are not engaging enough."
- User-Generated Solutions:
- 66.7% requested easier, one-click access to events.
- 66.7% advocated for gamification and sustainability challenges between faculties.
- 50% emphasized the need for a centralized digital platform and tangible rewards/incentives.

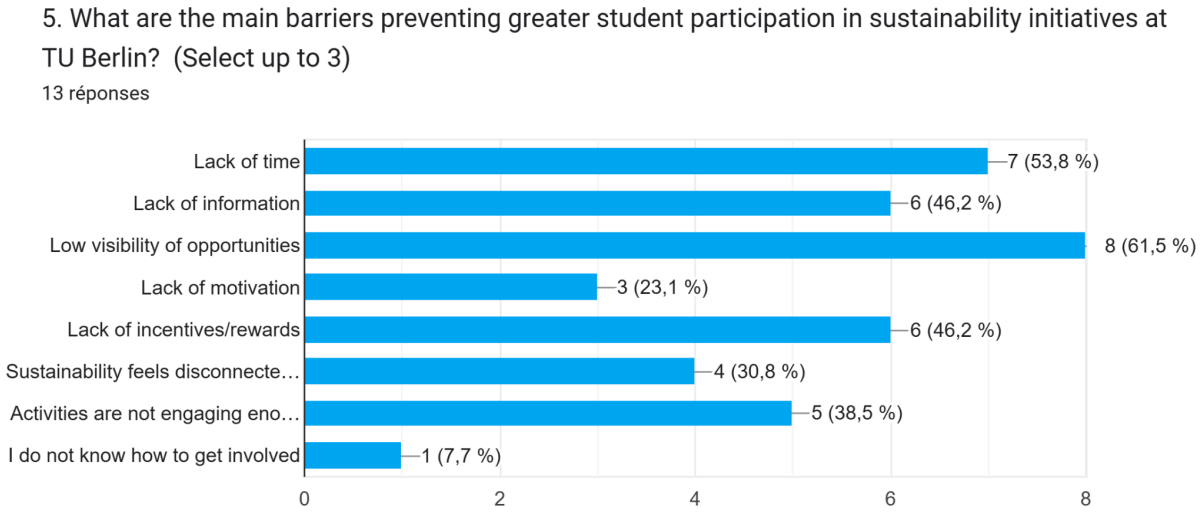


Figure 4. Main barriers to student participation in sustainability initiatives.

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The link of the survey:

https://docs.google.com/forms/d/e/1FAIpQLScX--M_X-ybX7qrg4v6PaztTbl4PMBY7FP9S7_lOJLzaa9Jlg/viewform?usp=dialog

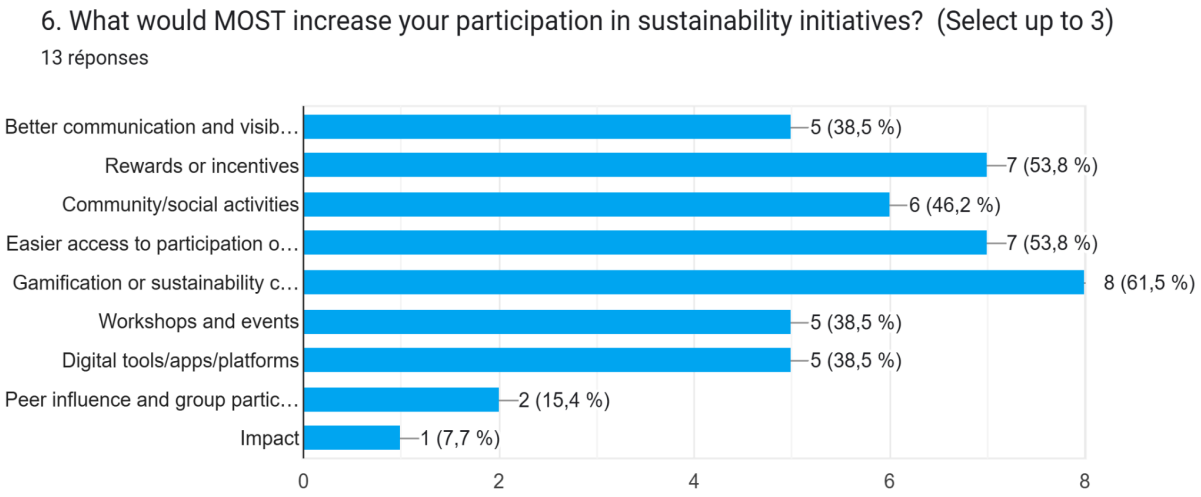


Figure 5. Factors that would increase students' participation in sustainability initiatives.

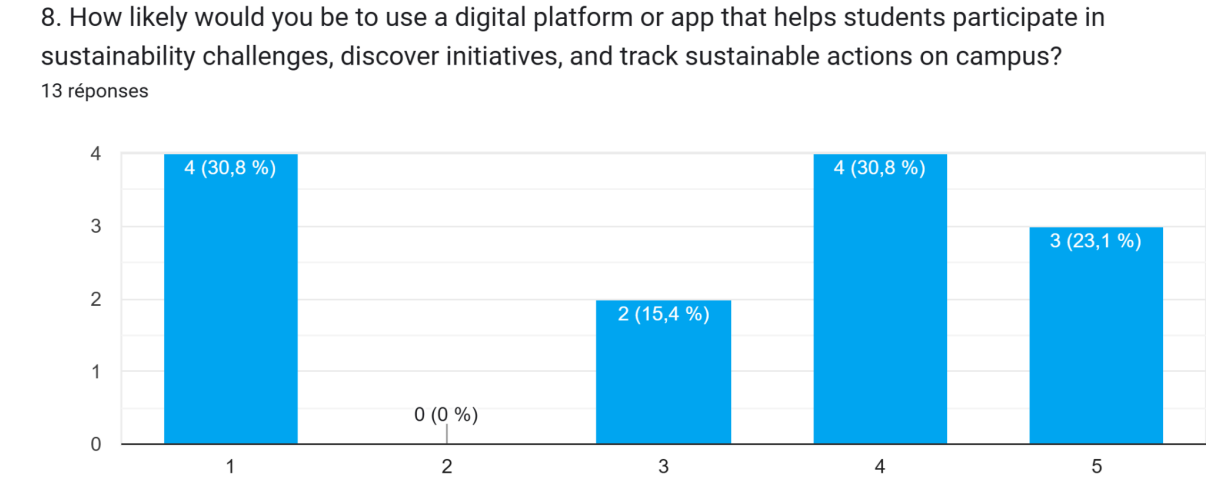


Figure 6. Bar chart showing students' likelihood of using a digital tool to help them participate in sustainability initiatives.

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4. The Engagement Toolkit: Tested Prototypes & Concepts

To address the documented barriers of **"Lack of Time"** and **"Low Visibility,"** which come out as a survey outcomes, we designed a modular engagement toolkit containing physical and digital prototypes:

4.1. Physical Prototypes (Tested at ECDF Hackathon)

Prototype A: The Belonging Map



Figure 7. The Belonging Map that was used by Hackathon members.

The Concept: A spatial, human-centered design tool where participants place colored pins/stickers on a high-resolution 3D campus map poster (*Figure* 7):

- 🔵 **Blue:** Places where they feel most connected or experience a sense of community ("I belong here").
- 🔴 **Red:** Places they rarely use or feel excluded from.
- 🟡 Yellow: Places they see potential for environmental or social improvement ("Needs improvement").

The Outcome: Participants added qualitative sticky notes explaining *why* they chose these zones. During the ECDF Hackathon, we observed heavy green/blue pinning around the Mensa and underutilized administrative red zones, highlighting strategic locations for future physical sustainability campaigns and events.

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Prototype B: Sustainability Bingo Together (Icebreaker)

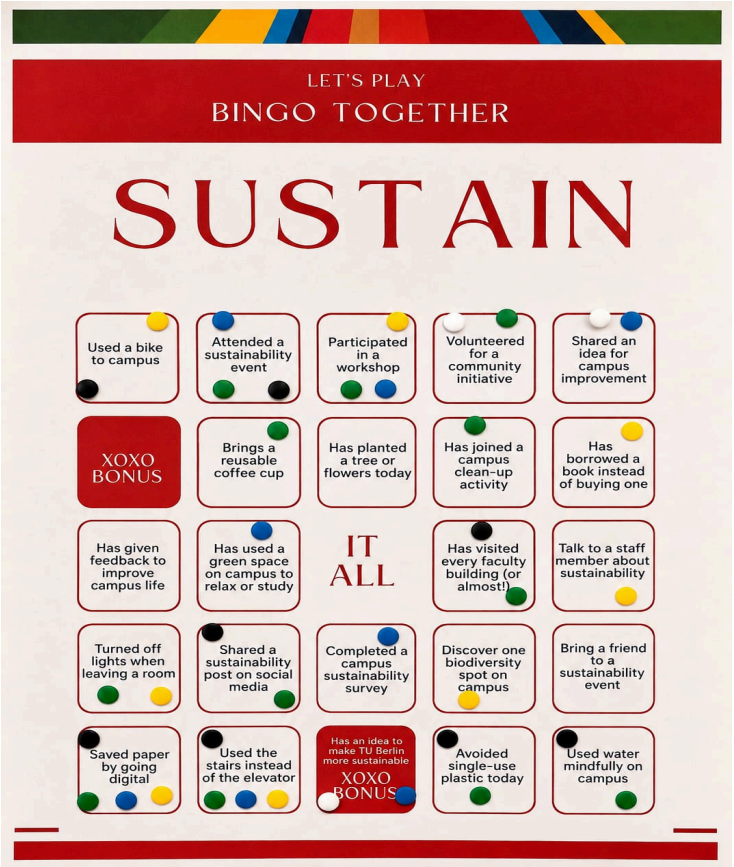


Figure 8. Let's Play Bingo Together

The Concept: A gamified, printed poster with a grid of daily sustainability actions (e.g., *"Used a bike to campus," "Brings a reusable coffee cup," "Has given feedback to improve campus life").*

The Outcome: Implemented at ECDF as a large bulletin board. Participants pinned their names to actions they completed. This lowered emotional barriers, verified existing pro-environmental behaviors, and initiated active dialogue among stakeholders.

4.2. Digital Prototypes (TU Sustainability Hub App)

To address the digital participation gap identified during the Sustain.ALL projects, we designed a high-quality application prototype called the TU Sustainability Hub App. The prototype was developed as a proof of concept to demonstrate how a future digital application could centralize sustainability participation at TU Berlin. It is not yet publicly deployed and has not been released as a website or mobile application.

The prototype includes 8 interconnected interfaces: Dashboard, Missions, Events, Proposals, Community, Rewards, Leaderboard, and Profile. These screens illustrate how users could discover sustainability opportunities, participate in campus activities, submit ideas, collaborate with peers, and receive recognition through a gamified participation system.

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4.2.1 Core platform features

The prototype is organized into eight interconnected sections: Dashboard, Missions, Events, Proposals, Community, Rewards, Leaderboard, and Profile. Together, these features support the complete participation journey from discovering activities to receiving recognition.

Feature	Purpose
Dashboard	Displays personal progress, XP level, CO_2 saved, completed missions, and upcoming activities.
Missions	Offers categorized sustainability challenges such as transport, lifestyle, workshops, volunteering, and co-creation.
Events	Allows users to discover and register for sustainability events and workshops.
Proposals	Enables students to submit sustainability ideas and support proposals from peers.
Community	Provides a discussion space for sharing ideas, forming clubs, and exchanging feedback.
Rewards	Converts earned XP into badges, vouchers, and sustainability recognition.
Leaderboard	Encourages friendly competition through faculty and campus rankings.
Profile	Tracks individual participation history and achievements.

Table 1. Eight interconnected sections

4.2.2 User participation workflow

The platform was designed around a simple participation cycle that encourages continuous engagement rather than one-time involvement.

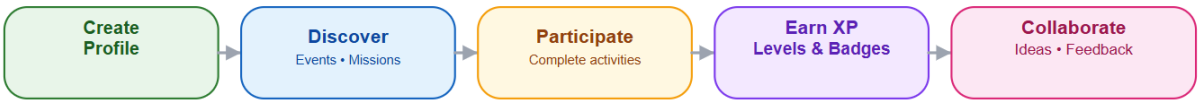


Figure 9. User journey within the TU Sustainability Hub prototype.

4.2.3 Gamification for sustainability engagement

Gamification was integrated into the TU Sustainability Hub prototype as a mechanism to encourage participation and provide continuous feedback on user engagement. Rather than turning sustainability into a game, these features were designed to recognize students'

contributions, visualize progress, and motivate long-term involvement in campus sustainability initiatives.

The participation model follows a simple progression:

Missions → XP → Levels → Badges → Rewards → Leaderboard

Users earn experience points (XP) by completing sustainability-related activities across different categories, including transport, lifestyle, volunteering, workshops, and co-creation.

Example missions include:

- ★ Bike to Campus (+30 XP)
- ★ Submit a Green Idea (+60 XP)
- ★ Meatless Monday Challenge (+80 XP)
- ★ Zero-Waste Week (+100 XP)
- ★ Co-Creation Workshop (+150 XP)

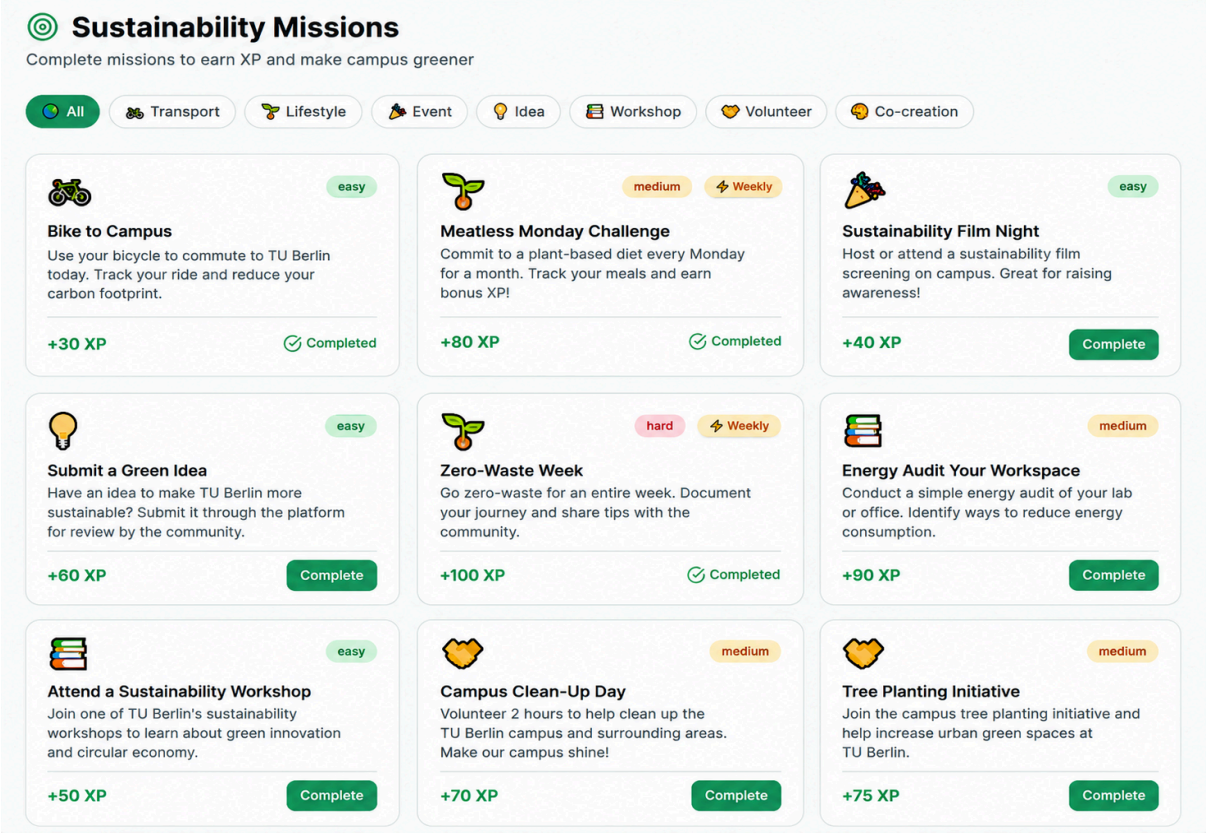


Figure 10. Sustainability Missions Interface

Accumulated XP contributes to user levels, digital badges, leaderboard rankings, and prototype rewards such as meal vouchers, extended library access, sustainability merchandise, and campus recognition. As this is a proof-of-concept prototype, the effectiveness of these gamification mechanisms would require further usability testing and evaluation with TU Berlin users.

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4.2.4 Student proposals and collective participation

A key feature that emerged during the hackathon was the proposal system. Rather than limiting participation to individual actions, the platform enables students to contribute ideas for improving campus sustainability and allows other users to support them collectively.

The proposed participation model follows a collaborative pathway:

Student submits an idea → Community supports the proposal → High-support proposals gain visibility for potential institutional review.

This mechanism creates a potential bridge between bottom-up student participation and university decision-making, strengthening democratic engagement within the campus sustainability ecosystem.

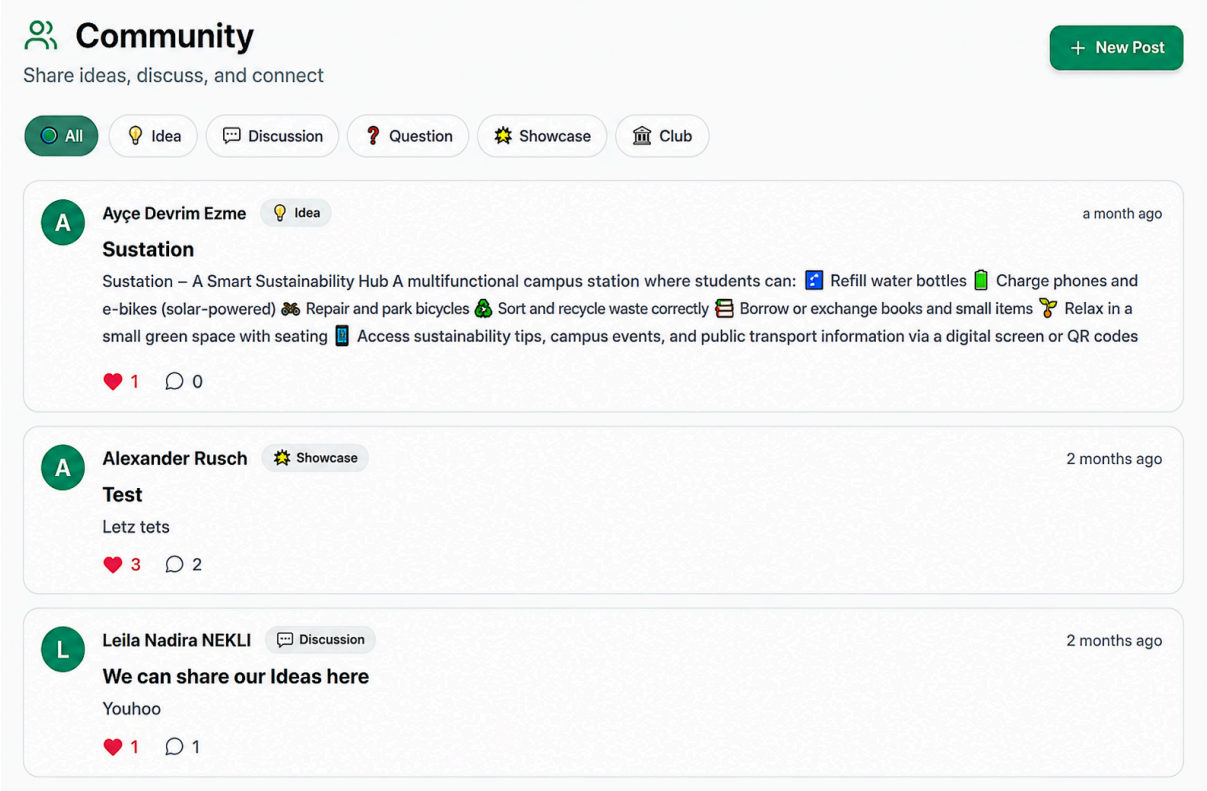


Figure 11. Community Interaction Interface

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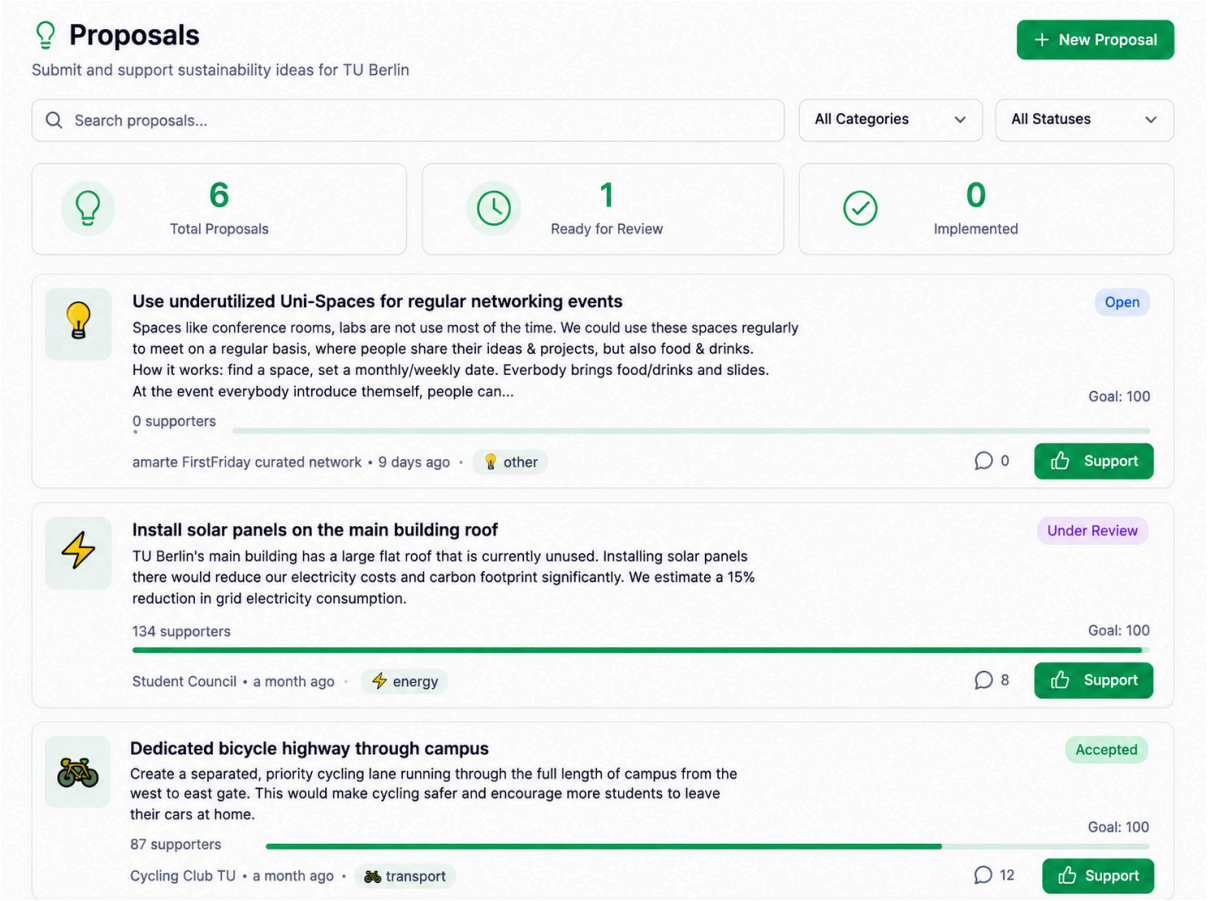


Figure 12. Student Sustainability Proposals

4.2.5 Prototype contribution

The TU Sustainability Hub App demonstrates how a single digital platform could integrate sustainability opportunities that are currently dispersed across different communication channels. Although the prototype has not yet been implemented at TU Berlin, it provides a practical model for increasing visibility, participation, collaboration, and recognition of sustainability engagement. Future work should evaluate its usability and its effectiveness in influencing long-term participation behavior.

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4.2.6 TU Sustainability Hub interface

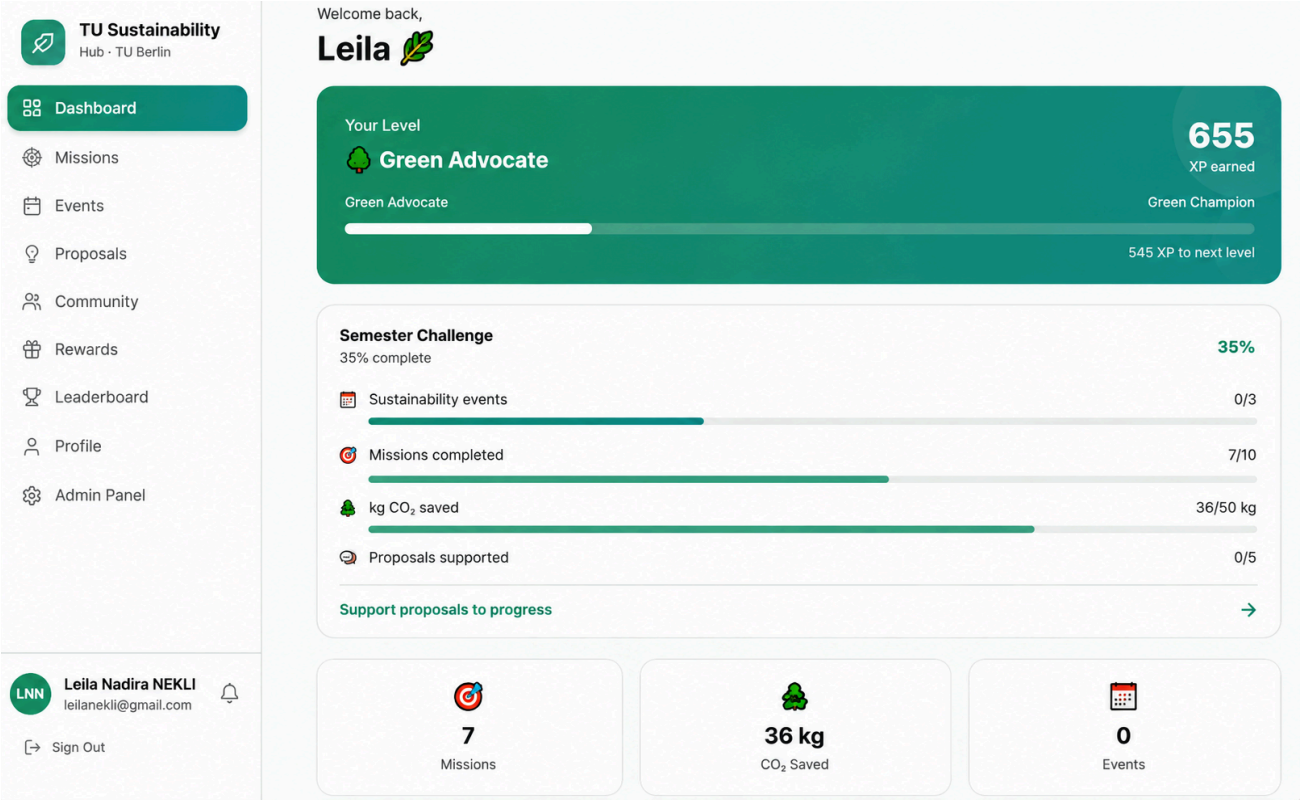


Figure 13. Dashboard Interface

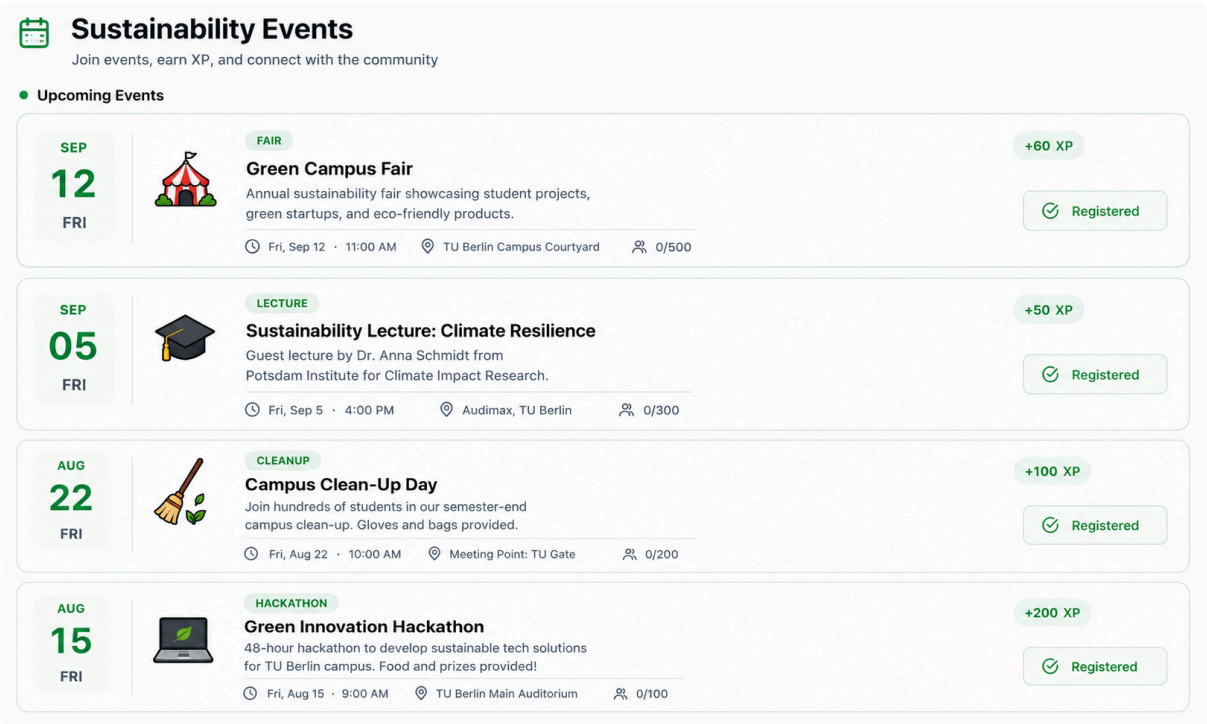


Figure 14. Sustainability Events Interface.

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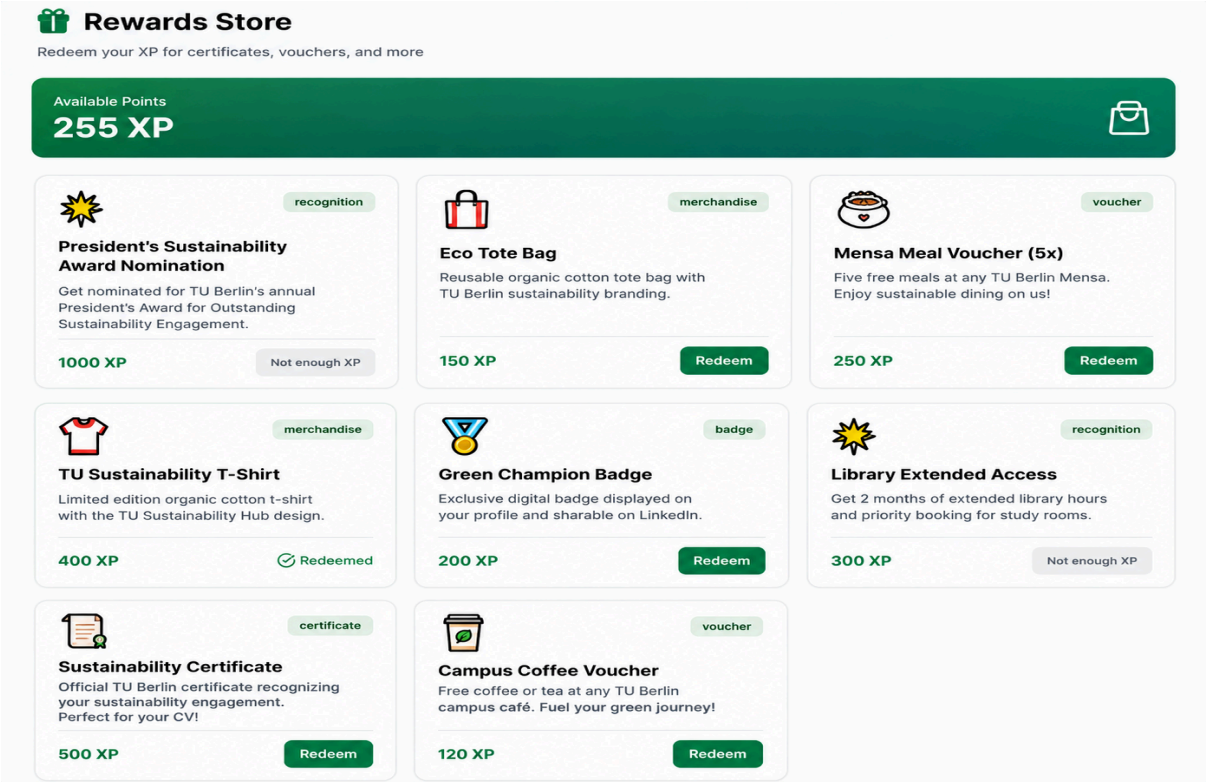


Figure 15. Rewards Store Interface

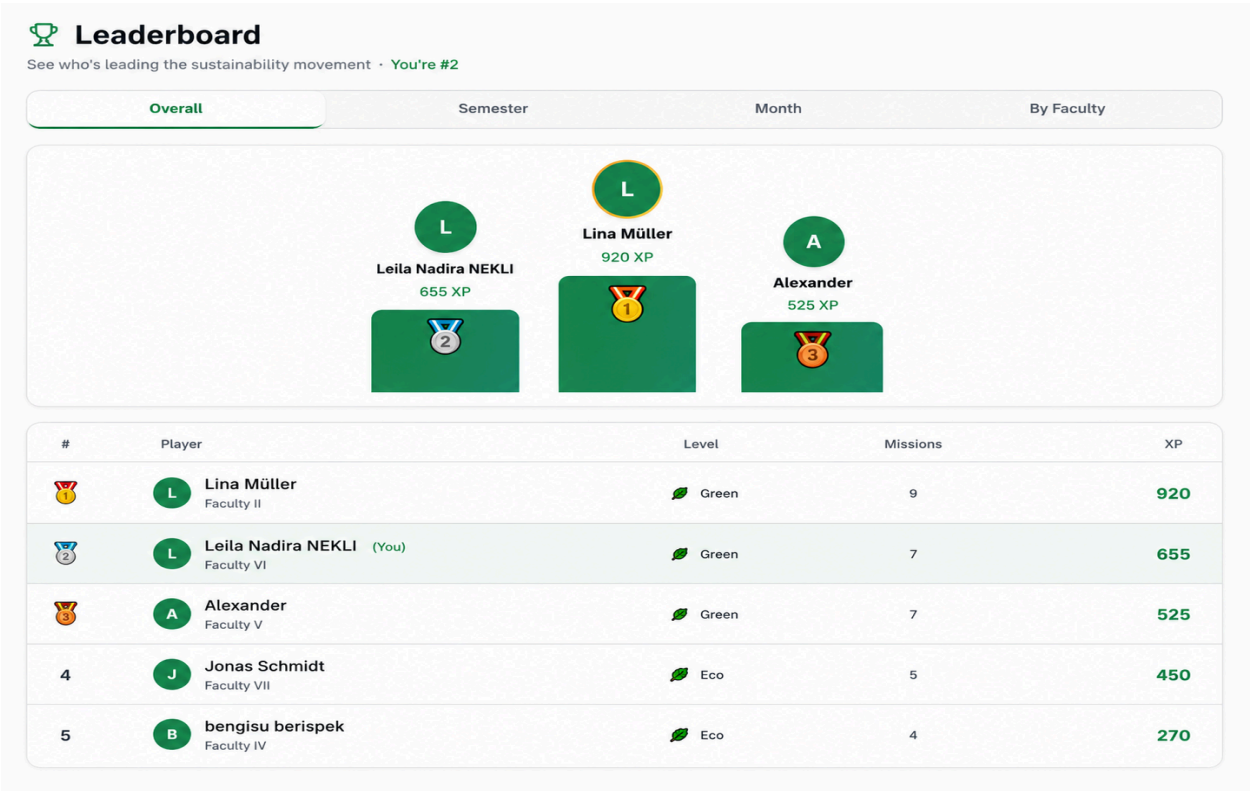


Figure 16. Leaderboard and User Ranking Interface

App Link: [TU Sustainability Hub]

5. Discussion and Critical Analysis

The project demonstrates that strengthening sustainability participation requires more than creating additional environmental initiatives. Our research indicates that visibility, accessibility, communication, and opportunities for collaboration are equally important factors influencing student engagement. By combining literature review, benchmarking, stakeholder and system mapping, the Belonging Map, the student survey, and co-creation activities, we developed the TU Sustainability Hub as an evidence-based response to the participation gaps identified throughout the semester.

A particularly important outcome of the project was the integration of co-creation into the design process. Feedback received during the Sustain.ALL Hackathon directly influenced the evolution of the prototype, including the introduction of the student proposal feature, demonstrating how stakeholder participation can improve the design of digital sustainability tools.

6. Project Limitations & Future Research

To ensure a balanced assessment, we consider both the structural limitations of our research and its potential for future development.

6.1. Project Limitations

- **Small Survey Sample :** With only 13 student responses, our survey data provides valuable qualitative indicators and behavioral trends, but it cannot be considered statistically representative of the entire TU Berlin student population.
- **Proof of Concept Prototype Stage:** The TU Sustainability Hub App is currently a non-deployed, high-fidelity visual mockup. It is a proof of concept rather than a fully implemented, functional IT platform.
- **Lack of Staff and Administrative Validation:** Due to strict data privacy (GDPR) regulations and administrative barriers, our surveys and prototypes were primarily tested with students. Direct input and validation from administrative staff remain outstanding.
- **Technical considerations:** Data security, user authentication, and responsible management of participation data require further development before institutional deployment.

6.2.Future Research

Future work should focus on a semester-long pilot implementation of the TU Sustainability Hub app. This would enable researchers to evaluate usability, user retention, participation patterns, and the effectiveness of gamification and community-driven proposals in supporting long-term sustainability engagement.

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7. Concrete Implementation Steps & Resource Requirements

The TU Sustainability Hub is proposed as a pilot implementation rather than an immediate university-wide platform. Based on this project's findings, the next step is to evaluate its usability, participation potential, and integration with existing sustainability initiatives at TU Berlin before broader institutional adoption.

To support this process, the implementation plan is structured into three phases, focusing on pilot development, beta testing, and institutional embedding.

7.1. **Phase 1:** Pilot & App Development (Months 1–3)

Objective: To develop the core version of the TU Sustainability Hub, test its concept with students and staff, and collect initial feedback on user engagement and platform usability.

Steps:

1. **Complete user testing**
   Publish the completed student survey to Faculty VI and Faculty IV to gather further feedback on student needs and the app's gamification concept.
2. **Develop the core platform**
   Develop the app's frontend and backend in collaboration with software engineering students from Faculty IV, focusing on the core features of the prototype, including events, sustainability missions, XP and badges, proposals, and community interaction.
3. **Create low-barrier participation channels**
   Place physical suggestion boxes at Mensa entrances to provide an analogue entry point for students who may not immediately engage with the digital platform and to collect additional feedback.

Resources Needed:

- **Budget:** Approximately €150 for printed promotional materials, feedback cards, and physical suggestion boxes.
- **Personnel:** Project leads coordinating with Sustain.ALL members and Faculty IV software engineering students.
- **Partnerships:** Faculty IV and Mensa Management (Studierendenwerk Berlin).

7.2. **Phase 2:** Beta Testing & Digital Platform Launch (Months 4–6)

Objective: Test the platform with selected student groups, improve its usability, and prepare the infrastructure required for a wider launch.

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Steps:

1. **Conduct closed beta testing**
   Test the TU Sustainability Hub with selected student groups from Faculty VI and Faculty IV to identify usability issues and improve the UI/UX.
2. **Explore Single Sign-On integration**
   Integrate the platform with TU Berlin's existing Single Sign-On (SSO) system to enable secure and accessible registration for students.
3. **Launch the beta version**
   Make the beta version available through the Google Play Store and Apple App Store, supported by a targeted digital communication campaign.

Resources Needed:

- **Technical Infrastructure:** Integration with TU Berlin's existing SSO and IT systems, together with secure cloud hosting.
- **Budget:** Approximately €6,000 (€5,000 for backend development and cloud infrastructure and €1,000 for promotional materials and beta-tester incentives).
- **Personnel:** 2 student developers from Faculty IV and 1 UI/UX designer.
- **Partnerships:** TU IT Services, Faculty IV, and the Einstein Center Digital Future (ECDF).

7.3. **Phase 3:** Evaluation, Administrative Embedding & Scaling (Months 7–12)

Objective: Evaluate the pilot, connect the platform with existing sustainability structures, and assess its potential for wider implementation across TU Berlin.

Steps:

1. **Evaluate the pilot**
   Examine user participation frequency, feature usage, participation in sustainability missions and events, engagement with the proposal function, and user feedback on usability and overall experience.
2. **Connect the platform with existing sustainability reporting**

   Explore how participation data generated by the platform could support future sustainability monitoring and reporting within Office K-7 Sustainability.

3. **Introduce university-wide engagement mechanisms**
   Test an inter-faculty Sustainability Championship within the app to encourage continued participation and explore the potential of gamification at a larger scale.
4. **Establish long-term funding and partnerships**
   Develop a sustainable funding model for platform maintenance and student rewards, including potential institutional funding and external partnerships.

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Resources Needed:

- **Budget:** Approximately €2,500 annually for reward incentives and platform maintenance
- **Personnel:** 1 part-time campaign manager responsible for coordinating the platform and its institutional partnerships.
- **Partnerships:** The Sustainability Office, Office K-7, relevant faculties, student initiatives, and other institutional stakeholders.

The pilot would provide the evidence needed to assess whether the TU Sustainability Hub can effectively support student participation and complement existing sustainability initiatives. Rather than assuming immediate university-wide adoption, the phased approach allows the technical, organizational, and engagement aspects of the platform to be tested and adapted based on actual user experience.

Estimated Pilot Budget

Item	**Estimated Cost (€)**
Printed promotional materials, feedback cards & suggestion boxes	€150
Student/software development & cloud infrastructure	€5,000
Beta launch, promotional materials & tester incentives	€1,000
Rewards & platform maintenance	€2,500 / year
Estimated First-Year Total	**€8,650**

Table 2. Pilot Project Cost Estimation Table

Budget: Approximately €6,000 (€5,000 for student development support, backend development, and technical infrastructure, and €1,000 for UI/UX improvements, beta-testing support, launch communication, and tester incentives). The development estimate is informed by current app-development rates in Berlin **[12].**

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8. Reflection & Strategic Suggestions for the Next Cohort

Future Research and Development

A potential next step would be to test the platform with TU Berlin students and staff during a future semester.

The testing could investigate:

- How frequently users participate.
- Which features they use.
- Which missions and activities attract participation.
- Whether gamification encourages continued engagement.
- Which features should be improved or removed.
- How students use the proposal function.
- How the platform could connect with existing university sustainability activities.

The results could then be used to refine the platform and evaluate whether the concept has potential for broader implementation.

9. Conclusion

This project investigated how sustainability participation at TU Berlin can be strengthened through stakeholder engagement, co-creation, and digital participation tools. Rather than beginning with a predefined solution, we first sought to understand student's behavior, sense of belonging, participation barriers, and existing sustainability practices through literature review, benchmarking, stakeholder and system mapping, the Belonging Map, and a student survey. These research activities informed every stage of the project.

The findings highlighted that participation is influenced not only by motivation but also by the visibility, accessibility, and coordination of sustainability opportunities. The hackathon and co-creation process further demonstrated the value of involving students in developing solutions and directly influenced the evolution of our prototype.

As the practical outcome of this research, the TU Sustainability Hub proposes a centralized digital platform that brings together sustainability events, missions, workshops, community interaction, student proposals, and gamification within a single participation ecosystem. Although it remains a proof-of-concept prototype, it provides a realistic foundation for future pilot testing, refinement, and institutional development.

Ultimately, this project demonstrates that achieving sustainability transformation at TU Berlin requires not only ambitious environmental goals but also inclusive mechanisms that enable students and staff to participate, collaborate, and contribute to long-term behavioral change.

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